Passive ISAC for Urban Transit using Self-Supervised World Models

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***Abstract*—Integrated Sensing and Communications (ISAC) enables the opportunistic reuse of existing wireless infrastructure, allowing commodity Wi-Fi access points to passively classify passenger mobility dynamics in public transport without dedicated radar or optical sensors. While Received Signal Strength Indicator (RSSI) sequences are natively accessible across off-the-shelf devices, non-stationary multi-device interference severely limits classical supervised classifiers, which previously reached a Matthews Correlation Coefficient (MCC) of 0.756. To overcome these limitations, this paper introduces self-supervised Joint-Embedding Predictive Architecture (JEPA) world models, regularized sequence baselines, and selective state-space models evaluated on a primary 3,511-sample transit dataset. On this benchmark, Time-Series JEPA (TS-JEPA) achieves an MCC of 0.9194, while a 16-member stacked ensemble reaches 0.9232 (0.9563 under multi-device noise) and a compact three-member self-supervised subset attains 0.9160 with only 3.42 million parameters. Crucially, self-supervised representations generalize robustly across distinct collection regimes where supervised baselines experience catastrophic collapse (dropping from 0.8858 to 0.4872). Channel-level Shapley attribution and architecture-controlled ablation confirm that cross-domain feature synergy drives general-purpose world models, while dedicated lightweight models like LeJEPA deliver near-peak accuracy (0.9143 MCC) directly over 4 base kinematic channels for edge deployment.**

***Index Terms*—ISAC, Wi-Fi sensing, self-supervised learning, JEPA, RSSI, passenger classification, state space models, ensemble methods**

I. Introduction

Urban public transportation networks require continuous passenger mobility monitoring to optimize fleet dispatching and route occupancy. While conventional automated passenger counting relies on optical cameras or infrared beams, these solutions incur substantial maintenance costs and raise privacy concerns. In contrast, opportunistic Integrated Sensing and Communications (ISAC) transforms standard wireless infrastructure into dual-functional sensing nodes without dedicated radar hardware. Commodity onboard Wi-Fi access points passively capture Received Signal Strength Indicator (RSSI) streams from client devices during routine network exchanges, enabling non-intrusive passenger monitoring without optical surveillance or dedicated radar deployment. While opportunistic radio sensing avoids recording visual biometrics or explicit user credentials, it presents subtle security and privacy considerations: continuous radio signal trajectories can inadvertently expose passenger mobility patterns to machine-learning profiling, even when individual device identifiers are anonymized.

In transit monitoring, mobility is cast as classifying four operational states: internal vehicle transit, alighting transitions, boarding events, and stationary platform waiting. Prior research established an initial benchmark of 2,251 RSSI samples [1], where a Gaussian Process achieved an Matthews Correlation Coefficient (MCC) of 0.756 [2]. However, supervised neural networks face severe limitations in vehicular radio environments: RSSI streams suffer from multipath fading and cross-traffic interference from concurrent devices, and supervised classifiers trained on limited annotations tend to memorize antenna-specific interference artifacts rather than underlying transit kinetics, resulting in brittle boundaries under environmental drift. To overcome this fragility, self-supervised learning via Joint-Embedding Predictive Architecture (JEPA) world models provides a compelling alternative: rather than reconstructing noisy high-frequency signals, JEPA models predict abstract representations in a latent embedding space, capturing physical motion constraints while discarding non-semantic radio noise.

In this paper, we address these challenges through four contributions. First, we formulate passenger mobility sensing within a self-supervised world model framework, deploying four distinct JEPA architectures (Time-Series Joint-Embedding Predictive Architecture (TS-JEPA), LeJEPA, CF-JEPA, T-JEPA) alongside selective state-space models and regularized baselines. Second, we introduce a primary 3,511-sample transit dataset and quantify stability under collection expansion, where self-supervised world models improve as the collection grows (+0.042 MCC) while every deep supervised baseline degrades. Third, we construct an 18-channel hierarchy, evaluate attribution via SHapley Additive exPlanations (SHAP), and demonstrate through architecture-controlled ablation that cross-domain feature interaction drives multi-channel tokenizers, while dedicated compact models (LeJEPA at 0.566 M parameters) achieve 0.9143 MCC directly over 4 base channels. Fourth, we develop temperature-calibrated ensembles, achieving a peak MCC of 0.9232 and isolating a compact three-member self-supervised ensemble delivering 0.9160 MCC with 3.42 M parameters.

The remainder of this paper is organized as follows. Section II reviews related work, and Section III details the setup, feature hierarchy, JEPA world models, and calibrated stacking. Section IV analyzes empirical classification benchmarks,

collection-expansion stability, feature attribution, and edge trade-offs, and Section V concludes the paper.

II. Related Work

A. Passive Wi-Fi Sensing and Passenger Monitoring

Automatic passenger counting is essential in public transit. While video-based Automatic Passenger Counting (APC) systems deploy cameras for neural tracking [3], they require dedicated hardware and raise severe privacy concerns. In contrast, Wi-Fi sensing passively monitors radio emissions from passenger devices at zero hardware expense. While Channel State Information (CSI) provides subcarrier resolution for cabin monitoring [4], it requires custom firmware. The RSSI, by contrast, is ubiquitous across standard routers [5], enabling localization [6], probe occupancy counting [7], signature tracking [8], and activity classification [9].

B. Integrated Sensing and Communications (ISAC)

Integrated sensing and communications leverages cellular and wireless infrastructure as dual-functional sensing nodes [10], [11], where machine learning extracts physical environment states from ambient signals [12]. Our work adopts this opportunistic paradigm, classifying transit states from RSSI time series using standard onboard access points without dedicated radar units.

C. Sequence Modeling and World Models

Sequence modeling has advanced through selective state spaces (Mamba [13], Mamba-3 [14]) and self-supervised joint-embedding architectures (I-JEPA [15], LeJEPA [16]). Recent extensions adapt JEPAs to tabular and temporal data [17]–[19], offering strong potential for radio sensing where labeled traces are scarce [20]. While prior benchmarks achieved 0.756 MCC via Gaussian Processes [2], this work is the first to deploy self-supervised world models for opportunistic Wi-Fi ISAC in public transit.

D. Security, Privacy, and Profiling in Wireless Sensing

Although passive radio sensing avoids recording visual or acoustic biometrics, opportunistic wireless sensing introduces distinct privacy vulnerabilities. Even when mobile operating systems enforce MAC address randomization, physical-layer radio characteristics, transmission burst timings, and sequential RSSI trajectories remain susceptible to device fingerprinting and behavioral tracking. Recent studies underscore that standard obfuscation techniques can be systematically defeated by machine-learning-driven profiling. Specifically, Silva et al. [21] demonstrated that user profiling models can extract consistent behavioral patterns and privacy preferences from observational interaction data, effectively re-identifying users despite privacy-preserving software abstractions. Furthermore, investigating the efficacy of differential privacy against profiling attacks, Silva et al. [22] established that even mathematically grounded noise addition can be circumvented when machine-learning classifiers exploit aggregate temporal correlations and structural invariants. These findings highlight that opportunistic ISAC cannot rely solely on identifier anonymization or naive perturbation, requiring conscious architectural isolation and privacy-preserving representation learning.

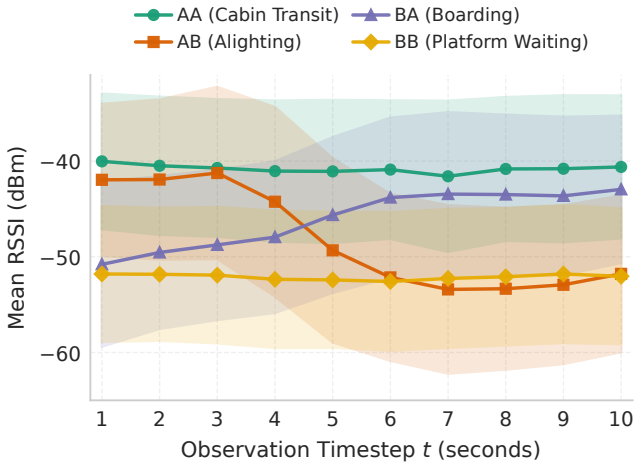


Figure 1. Mean RSSI trajectory per passenger movement class ($\pm 1\,\sigma$ shaded variance band) across 10-second observation windows on the 3,511-sample collection.

III. Materials and Methods

We formulate passenger movement classification as a four-class time-series task over short Wi-Fi RSSI windows. This section details the physical collection environment, the 18-channel feature pipeline, the self-supervised world models, the supervised baselines, and the calibrated ensemble stacking framework.

A. Physical Sensing Setup and Dataset Composition

The physical sensing environment comprises two spatial zones emulating public transport interactions as established in [2]: Zone A denotes the interior passenger cabin (an enclosed room simulating a vehicle interior with a Wi-Fi access point adjacent to the doorway), and Zone B denotes the exterior platform or bus stop (the adjacent corridor). An onboard passive sniffer records 802.11 transmissions from passenger mobile devices. Spatial transitions define four mutually exclusive movement classes: internal transit (AA, Zone A $\rightarrow$ Zone A), alighting (AB, Zone A $\rightarrow$ Zone B), boarding (BA, Zone B $\rightarrow$ Zone A), and platform waiting (BB, Zone B $\rightarrow$ Zone B).

Observations consist of 10-second RSSI sequences sampled at 1 Hz ($T = 10$, $t \in \{1, \ldots, 10\}$ seconds). As depicted in Figure 1, each class exhibits a characteristic radio attenuation profile: internal transit (AA) maintains a high, stable signal (≈ -40 dBm), alighting (AB) exhibits a monotonic decline from -42 dBm to -52 dBm, boarding (BA) rises from -51 dBm to -43 dBm, and platform waiting (BB) remains attenuated at -52 dBm. The shaded $\pm 1\,\sigma$ variance bands capture multi-device concurrent interference while confirming the physical separability of the mobility states.

The dataset contains 3,511 labeled sequences, divided into 160 single-device calibration traces (pure scenario, 40 per class) and 3,351 multi-device concurrent interference traces (noise scenario, ≈ 840 per class). Data collection follows the controlled indoor methodology of [2], scaling the hardware diversity from 4 smartphones in [2] to 7 distinct smartphone models across multiple vendors and Android releases. The 3,511-sample benchmark is a strict superset of the 2,251-sample collection in [1] under identical sensing protocols: it retains all eight devices of the earlier campaign together with their traces, and adds five further devices. Consequently, 74% of the expanded collection originates from devices already represented in the prior one, and comparisons between the two quantify stability under collection expansion rather than transfer to an independent environment.

Data is partitioned with an 80/20 class-stratified split, holding out 12.5% of training data for validation ($\approx 10\%$ of total data). Training folds are augmented within folds using class-conditioned Gaussian noise ($\sigma = 0.05$) and intra-class Mixup ($\alpha = 0.4$), yielding 35,839 training samples with zero split contamination. Model selection is based on validation MCC, and the test split is evaluated once for final reporting.

B. 18-Channel Rich Feature Pipeline

To capture the temporal, statistical, and spectral dynamics of radio attenuation, we construct an 18-channel feature hierarchy from raw RSSI $r = [r_1, \ldots, r_{10}]$. The pipeline comprises four functional domains: kinematics and deviation (4 channels: raw r_t, velocity $\Delta_t = r_t - r_{t-1}$, acceleration $\Delta_t^2 = \Delta_t - \Delta_{t-1}$, and window deviation $d_t = r_t - \mu_w$); rolling statistical moments (4 channels: rolling mean $\mu_{t,w}$, standard deviation $\sigma_{t,w}$, dynamic range $R_{t,w}$, skewness $\gamma_{t,w}$, kurtosis $\kappa_{t,w}$, and velocity sign changes); spectral descriptors (4 channels: FFT DC power band_dc, AC residual energy band_ac, and peak location index); and time-frequency STFT sub-bands (4 channels: Hann-windowed sub-bands stft_b0 through stft_b3 capturing micro-Doppler motion). The plain baseline uses only the 4 kinematic channels, whereas the rich pipeline uses all 18 channels.

C. Self-Supervised World Models

All JEPA models learn invariant latent dynamics by predicting representations of masked or future temporal regions. An online context encoder f_θ is updated via gradient descent, a target encoder g_ξ is updated via Exponential Moving Average (EMA) ($\xi \leftarrow \tau\xi + (1-\tau)\theta$), and a latent predictor h_ψ minimizes the L2 prediction loss $\mathcal{L}_{\text{JEPA}} = \|h_\psi(f_\theta(x_c)) - \text{sg}(g_\xi(x_t))\|_2^2$, where $\text{sg}(\cdot)$ is the stop-gradient operator. We evaluate four distinct formulations, whose architectural parameters and pretraining settings are detailed in Table I:

T-JEPA [17] applies per-timestep channel tokenization with learned feature embeddings and two learnable `[REG]` tokens (prepended to context and target to prevent representational collapse), applying contiguous 30% temporal masks. **Time-Series JEPA (TS-JEPA)** [18] utilizes a 1D convolutional patch tokenizer (patch size two) with patch masking annealed from 0.4 to 0.7 across training. **LeJEPA** [16] executes next-step latent prediction without an EMA target, stabilized via Sketched Isotropic Gaussian Regularization (SIGReg) on the empirical characteristic function of latents. **CF-JEPA** [19] executes mask-free multi-horizon prediction across horizons $k \in \{1, 2, 3\}$ from context crops with an EMA target.

Pretraining runs for 500 epochs with AdamW, cosine annealing, gradient clipping at 1.0, and target momentum schedules shown in Table I. Fine-tuning employs progressive unfreezing (frozen backbone for 25% of epochs, then unfrozen at 0.1 learning rate with Mixup $\alpha = 0.4$ and label smoothing $\varepsilon = 0.1$). All fine-tuned models share a learned-attention pooling head mapping encoder latents through a two-layer MLP with dropout (0.3) to 4-class logits.

D. Supervised Classifiers and State-Space Baselines

SIGReg [16] is a 3-block 1D convolutional network (192 filters) with global pooling and 128-unit projection, regularized by SIGReg ($\lambda = 0.01$) over 256 random projections (0.697 M parameters, 4 channels). **Mamba-3 Sequence Models** [14] employ selective state spaces [13] (state dim 16, rank-4 MIMO) with four front-ends: Convolutional Neural Network (CNN) (0.945 M), Temporal Convolutional Network (TCN) [23] (0.843 M), Transformer (0.842 M), and multi-view fusion (1.701 M) over 4 channels. **DeepStack** is a two-stage stacked ensemble [24] combining ten base models via Level-2 networks and a DeepMetaMLP (43.576 M parameters). Supervised sequence baselines are tuned via Optuna [25] with Tree-structured Parzen Estimator (TPE) over 50 trials per model.

E. Calibrated Stacking and Pareto Optimization

To exploit complementary representations across model families while controlling edge complexity, we construct a calibrated stacking framework. Uncalibrated softmax outputs can produce overconfident misclassifications under noise. We therefore temperature-calibrate the raw logits z_m of each candidate model m by optimizing a scalar temperature parameter $T_m > 0$ on validation data to minimize Negative Log-Likelihood (Negative Log-Likelihood (NLL)) [26]:

$$p_m = \text{softmax}(z_m / T_m). \tag{1}$$

The calibrated class probability vectors are concatenated into a meta-feature vector $\mathbf{u} = [p_1^\top, \ldots, p_M^\top]^\top$ and passed to a multinomial logistic regression meta-learner [24] with L2 regularization penalty $C \in [0.005, 10.0]$.

Hyperparameter optimization for model selection and ensemble pruning is formulated as a bi-objective search over the Pareto frontier balancing classification MCC against total parameter count. Starting from an empty ensemble, a greedy Pareto algorithm iteratively evaluates candidate additions, selecting model subsets that maximize incremental MCC gain per added parameter. This directly yields minimal-footprint configurations tailored for edge devices.

Table I
ARCHITECTURAL SPECIFICATIONS AND PRETRAINING HYPERPARAMETERS ACROSS THE EVALUATED WORLD MODEL FAMILY.

Hyperparameter	TS-JEPA	LeJEPA	CF-JEPA	T-JEPA
Input Channels (C)	18 (Rich)	4 (Base)	18 (Rich)	18 (Rich)
Tokenizer Type	1D Conv Patch	Causal Temporal	1D Conv Patch	Per-feature + [REG]
Patch / Token Size	2 timesteps	1 timestep	2 timesteps	1 timestep
Embedding Dim (d)	256	128	256	128
Attention Heads (h)	8	4	8	4
Encoder Layers (L)	4	4	3	4
Feedforward Dim	512	256	256	256
Predictor Architecture	2-layer ($d = 128$)	3-layer ($d = 128$)	1-layer ($d = 128$)	3-layer ($d = 64$)
Predictive Objective	Masked patch L2	Next-step L2	Multi-horizon L2	Masked token L2
Anti-Collapse Mechanism	EMA target	SIGReg ($\lambda = 0.05$)	EMA target	EMA target + [REG]
Target EMA Schedule	$0.99 \rightarrow 0.996$	None (Direct)	$0.996 \rightarrow 0.999$	$0.99 \rightarrow 0.996$
Pretrain Epochs / Batch	500 / 128	500 / 128	500 / 128	500 / 128
Parameter Count	2.153 M	0.566 M	1.232 M	0.633 M

F. Channel-Level Interpretability & Attribution

To quantify the relative contribution of each channel within the 18-channel hierarchy, we employ Shapley Additive exPlanations (SHAP) [27]. Grounded in cooperative game theory, SHAP evaluates the marginal contribution of each feature channel j across all possible feature subsets (coalitions) $S \subseteq F \setminus \{j\}$:

$$\phi_j(x) = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} \left[f(S \cup \{j\}) - f(S) \right], \tag{2}$$

where F denotes the set of $M = 18$ feature channels, and $f(S)$ denotes model predictions conditioned on coalition S with absent features evaluated against empirical background distributions. The resulting attributions satisfy the efficiency axiom $\sum_{j=1}^{|F|} \phi_j(x) = f(x) - \mathbb{E}[f(x)]$, providing an exact decomposition of model predictions across input channels.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

This section evaluates the self-supervised world models, supervised sequence baselines, and calibrated ensembles across both experimental datasets, followed by feature channel attribution, architectural trade-offs, and privacy considerations.

A. Single-Model Classification Performance

Models were evaluated across both dataset collections. Table II reports the test MCC of each model family on the prior 2,251-sample dataset and on the new 3,511-sample dataset, along with the performance differential Δ. On the new collection, the four self-supervised world models achieve the top results: TS-JEPA reaches an MCC of 0.9194, LeJEPA obtains 0.9143, CF-JEPA achieves 0.9083, and T-JEPA attains 0.8982. Every evaluated JEPA architecture substantially exceeds the 0.756 classical Gaussian Process baseline [2].

By contrast, the selective state-space Mamba-3 TCN architecture scores 0.7816, and the DeepStack hierarchical ensemble reaches 0.7811. The supervised SIGReg classifier drops to 0.4872 despite leading the prior dataset at 0.8858, and the hyperparameter-optimized GRU baseline attains 0.5746. Every deep supervised architecture degrades as the collection expands, whereas the self-supervised world models improve, echoing the earlier finding that classical and supervised approaches are limited in this non-stationary radio environment [2] [1]. Figure 2 illustrates this behavior. Multi-seed evaluations across three independent runs (seeds 3, 5, 42) confirm protocol stability, with standard deviations constrained within narrow bounds ($\sigma \approx 0.008$ for SIGReg). Full pretraining over 300 to 500 epochs is mandatory: fine-tuning from early checkpoints (below 100 epochs) collapses test MCC to 0.49–0.60 across all seeds.

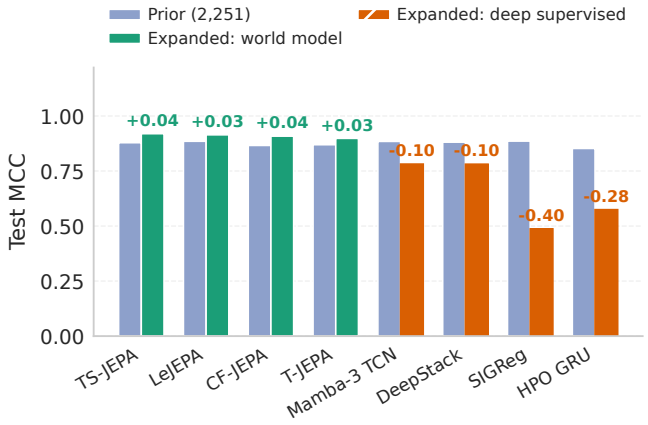


Figure 2. Collection-expansion comparison and single-model test MCC. Self-supervised world models gain as the collection expands, whereas deep supervised and state-space architectures degrade. Hatching denotes model family.

B. Stability Under Collection Expansion

The results in Figure 2 and Table II compare the prior 2,251-sample and expanded 3,511-sample collections. As noted in §III-A, the two collections share their device pool, so this comparison measures stability under collection expansion rather than generalization to an independent environment. All four self-supervised world models improve on the expanded collection,

Table II
SINGLE-MODEL TEST MCC ON THE PRIOR AND EXPANDED COLLECTIONS. Δ IS THE EXPANDED MINUS PRIOR DIFFERENCE.

Model	Prior	Expanded	Δ
TS-JEPA	0.8785	0.9194	+0.041
LeJEPA	0.8850	0.9143	+0.029
CF-JEPA	0.8661	0.9083	+0.042
T-JEPA	0.8694	0.8982	+0.029
Mamba-3 TCN	0.8844	0.7816	−0.10
DeepStack	0.8810	0.7811	−0.10
SIGReg	0.8858	0.4872	−0.40
HPO GRU	0.8524	0.5746	−0.28

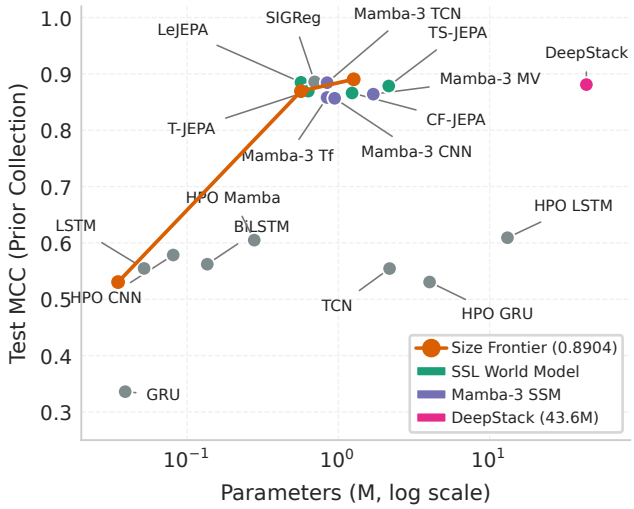


Figure 3. Parameter efficiency frontier. Each point denotes an individual model; the solid curve traces the greedy Pareto selection from logistic regression through LeJEPA to SIGReg.

gaining +0.029 to +0.042 in MCC, whereas the Mamba-3 TCN state-space model, the DeepStack ensemble, and the SIGReg classifier degrade.

The collapse of the supervised SIGReg classifier (-0.40 in MCC) exposes the vulnerability of deep supervised regularizers to a richer interference regime. Its characteristic-function regularizer penalizes deviations from an isotropic Gaussian latent prior conditioned strictly on the training labels; under more varied multi-device interference, this constraint creates brittle decision boundaries tailored to training artifacts. Self-supervised JEPA pretraining instead optimizes a predictive objective over radio sequence dynamics without label guidance, extracting latent motion invariants that are retained as the collection grows. This extends the earlier observation that supervised and classical learners plateau below 0.76 MCC in this environment [2]: the limitation is not annotation scarcity alone but the absence of a predictive objective over the radio signal itself.

Table III
ENSEMBLE CLASSIFICATION PERFORMANCE AND PARAMETER SCALING.

Ensemble	n	Params	Prior	New	Noise / Pure
21-member full stack	21	74.8 M	0.8873	**0.9232**	**0.9563** / **0.9596**
16-member subset	16	73.2 M	0.8873	**0.9232**	**0.9563** / 0.9179
World-only stack	7	6.68 M	0.8872	0.9065	0.9450 / **0.9596**
Best-world subset	3	3.42 M	0.8784	0.9160	0.9341 / 0.8773
Three-member Pareto	3	**1.26 M**	**0.8904**	0.5753	0.9396 / 0.9179

C. Calibrated Ensemble Performance and Pareto Efficiency

Table III summarizes ensemble performance and parameter scaling across datasets. Softmax probabilities are temperature-calibrated on validation NLL before fusion. On the prior collection, multinomial logistic regression stacking ($C = 0.005$) and entropy-weighted probability averaging both achieve an MCC of 0.8873, outperforming uniform probability averaging (0.8814) and Caruana selection (0.8706).

On the primary 3,511-sample collection, the 21-member calibrated logistic regression stack attains a peak test MCC of 0.9232, scoring 0.9563 on multi-device noise and 0.9596 on pure single-device traces. The 16-member non-redundant subset matches this peak performance (0.9232 MCC) while omitting weak recurrent backbones. For resource-constrained edge execution, the compact 3-member self-supervised subset (TS-JEPA, LeJEPA, and SIGReg) attains 0.9160 MCC with only 3.42 M parameters.

Figure 3 plots test MCC against parameter footprint across four orders of magnitude (0.039 M for GRU to 43.576 M for DeepStack). Self-supervised world models cluster tightly in an optimal high-performance zone between 0.566 M and 2.153 M parameters. While greedy Pareto selection on the prior collection identified a minimal 1.26 M parameter ensemble (logistic regression, LeJEPA, and SIGReg) achieving 0.8904 MCC, this specific subset drops to 0.5753 on the new collection due to the domain shift collapse of supervised SIGReg. In contrast, self-supervised world model ensembles deliver robust transfer across both environments (0.9065 to 0.9160 MCC), demonstrating near-peak accuracy at a fraction of the computational footprint.

D. Channel-Level Interpretability and Feature Ablation

To determine how each channel in the 18-channel hierarchy informs latent world-model predictions, we conduct additive feature attribution via SHAP on the primary dataset. Figure 4 reports the mean absolute Shapley value $|\phi_j|$ per channel evaluated on the leading TS-JEPA model. The attribution hierarchy reveals that macroscopic signal shape and low-frequency spectral power dominate the decision space. Peak location index ($|\phi| = 6.04 \times 10^{-3}$) and low-frequency STFT Band 0 (5.97×10^{-3}) contribute most strongly, capturing the exact timestamp of door transitions and the coarse Doppler shifts of physical motion. These are followed by rolling FFT DC power (4.40×10^{-3}) and AC energy (3.99×10^{-3}), which reflect baseline energy shifts between vehicle zones. In contrast, fine-grained statistical descriptors, such as rolling kurtosis

Table IV
ARCHITECTURE-CONTROLLED FEATURE CHANNEL ABLATION ACROSS WORLD MODELS ON THE PRIMARY DATASET (MCC).

Feature Configuration	C	TS-JEPA	CF-JEPA	T-JEPA
Full 18-Channel Hierarchy	18	**0.9194**	0.9083	0.8982
Top-12 SHAP Channels	12	0.5815	0.8098	0.5856
Base Kinematics (Raw, Δ, Δ^2, Dev)	4	0.4486	0.4366	0.4787
Rolling Statistics (Moments, Range)	5	0.3447	0.4351	0.4944
Spectral Domain (FFT, STFT 0–3)	6	0.2981	0.1869	0.2283
Raw RSSI Only (Single Channel)	1	0.2431	0.3305	0.3395

(1.22×10^{-3}), rolling standard deviation (0.95×10^{-3}), and high-frequency STFT bands, exhibit negligible marginal value, primarily capturing non-semantic multipath scattering rather than transit kinematics.

To validate these attribution findings under controlled conditions, Table IV presents an architecture-controlled feature ablation across TS-JEPA, CF-JEPA, and T-JEPA. When input channels are restricted to individual functional domains or the 4-channel base kinematic subset under identical model architectures, performance drops significantly (e.g., from 0.9194 to 0.4486 MCC for TS-JEPA on 4 channels, and to 0.2431 on raw RSSI alone). This confirms that self-supervised world models critically rely on cross-domain synergy across kinematic, spectral, and statistical features to construct noise-invariant latent representations under multi-device interference. By contrast, specialized lightweight architectures designed specifically for low-dimensional inputs, such as LeJEPA (0.566 M parameters with only 4 channels), achieve 0.9143 MCC, demonstrating that edge access points can choose between rich multi-channel tokenization and ultra-compact dedicated models with virtually zero performance loss.

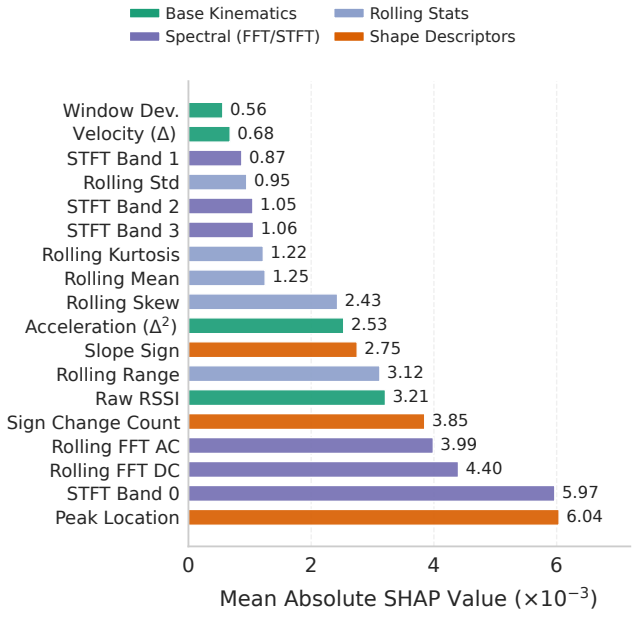


Figure 4. Channel-level SHAP feature attribution across the 18-channel hierarchy evaluated on the leading TS-JEPA world model.

E. Discussion: Architectural Insights and Sensing Trade-offs

The experimental findings offer three primary insights into passive ISAC for urban transit using self-supervised world models. First, the contrast between self-supervised world models and deep supervised baselines across collection scales reveals the mechanism of radio representation learning. In radio sensing, concurrent transmitting devices introduce non-stationary interference that alters local signal distributions. Deep classifiers regularized with strict isotropic Gaussian priors, such as SIGReg, constrain the latent space around training distributions; when unseen multi-device interference patterns appear, these rigid decision boundaries deteriorate, causing a 0.40 drop in test MCC. In contrast, self-supervised JEPA pretraining optimizes a predictive objective directly across masked time-frequency patches without label supervision, learning smooth, invariant latent trajectories corresponding to physical passenger displacement.

Second, the channel-level SHAP attribution and controlled ablation demonstrate that rich feature synergy is essential for general-purpose world models, since kinematic extrema, low-frequency Doppler power, and baseline energy shifts provide complementary physical constraints. Reducing input dimensions degrades fixed 18-channel patch tokenizers, whereas compact architectures like LeJEPA retain near-peak accuracy over 4 base channels, confirming flexible edge deployment options.

Third, the Pareto analysis demonstrates that massive neural ensembles are unnecessary for robust passive sensing. While the 21-member ensemble requires 74.8 M parameters, a compact three-member world model ensemble achieves 0.9160 test MCC at only 3.42 M parameters. To assess deployment viability beyond parameter count, we exported LeJEPA to int8 and executed it on an ESP32-D0WD-V3 (240 MHz, 4 MB flash, no PSRAM). The model requires 0.609 MiB at int8 and does not fit the 281 KiB of usable heap at any precision, so weights are memory-mapped from flash and read through the cache; the measured forward pass is 636.7 ms median with a 0 byte heap delta. This is a feasibility result, not a real-time one, as discussed below.

F. Privacy, Security, and Profiling Considerations

Deploying opportunistic ISAC in public transit environments requires balancing sensing utility against passenger privacy and security. While passive Wi-Fi sensing avoids recording identifiable optical imagery or acoustic signals, the omnidirectional broadcast nature of 802.11 transmissions creates a non-trivial attack surface. Unauthorized sniffers or compromised access points can intercept raw RSSI sequences and packet metadata without actively transmitting or alerting clients.

A primary privacy threat stems from machine-learning profiling. As established in Section II-D, profiling algorithms can circumvent identifier anonymization and differential privacy by exploiting temporal dependencies and structural sequence correlations [21], [22]. In vehicular transit this is amplified by the regularity of daily commutes: repeated transitions through cabin doorways produce characteristic attenuation signatures

which, combined with fixed route timetables, let an adversary link individual transit trajectories across days even under MAC address randomization.

Our architectural findings offer a practical, privacy-by-design mitigation through edge-confined inference. Because dedicated self-supervised architectures such as LeJEPA achieve near-peak performance (0.9143 MCC) with only 0.566 M parameters directly over 4 base kinematic channels (Table IV), classification can execute entirely within the onboard access point firmware. Under this paradigm, raw RSSI time series and intermediate latent representations are processed ephemerally in volatile memory and discarded immediately after state prediction; only aggregated, anonymized passenger flow counts (Δboarding, Δalighting) reach the transit management backend. Precluding the transmission and storage of continuous radio sequences prevents centralized repository harvesting and drastically restricts the window available for adversarial profiling.

G. Limitations and Deployment Outlook

Several constraints bound the scope of these results. The collection protocol used two controlled indoor zones rather than an operational vehicle, the device pool comprises thirteen smartphone models, and no participant demographic attributes were recorded, so the reported accuracy carries no guarantee of equivalence across device populations or passenger groups. Class labels were assigned by supervised session protocol rather than independent annotation, and the 160 single-device calibration traces are too few to support a separate pure-scenario held-out evaluation. We also note that the training corpus overlaps the evaluation splits, so the absolute values in Table II are optimistic; the comparisons across model families, drawn from identical splits under one protocol, are the claims the table supports.

Deployment raises further questions this study does not resolve. Single-access-point inference cannot disambiguate passengers moving through spatially overlapping zones, and cooperative multi-AP fusion with spatial diversity remains unmeasured. The int8 LeJEPA measurement of 636.7 ms per 10-second window on an ESP32-D0WD-V3 establishes feasibility but not real time, and it covers one model rather than an ensemble. Energy consumption, thermal behavior under continuous operation, and behaviour across multiple vehicles, routes, and seasonal radio drift are likewise outside the present evaluation.

V. Conclusion

This paper demonstrated opportunistic Integrated Sensing and Communications (ISAC) for passenger movement classification in public transport using commodity Wi-Fi infrastructure. Passively capturing standard RSSI sequences without dedicated radar or firmware modifications, self-supervised JEPA world models learn environment-invariant representations of passenger mobility. On the primary 3,511-sample dataset, TS-JEPA achieves a single-model MCC of 0.9194, while a 16-member calibrated stacked ensemble reaches 0.9232 (0.9563 on multi-device noise) and a compact 3-member self-supervised subset attains 0.9160 with 3.42 M parameters.

A central insight is the contrast in behavior between self-supervised world models and deep supervised baselines as the collection expands. While self-supervised world models improve across collection scales (+0.029 to +0.042 MCC), every deep supervised architecture degrades, with the regularized SIGReg baseline collapsing from 0.8858 to 0.4872. Because the expanded collection shares its device pool with the prior campaign, this contrast quantifies stability under collection expansion rather than transfer to an independent environment. Furthermore, game-theoretic SHAP attribution and controlled feature ablation reveal that cross-domain interaction between kinematic and spectral descriptors drives multi-channel models, while dedicated lightweight models like LeJEPA deliver near-peak performance (0.9143 MCC) directly over 4 base kinematic channels for edge execution. These findings establish that commodity access points can effectively serve as dual-functional sensing nodes in intelligent transit networks. Future work will explore cooperative multi-access-point spatial diversity and online continual adaptation for seasonal radio drift.

Furthermore, to address the privacy vulnerabilities identified in continuous radio sensing, future investigations will develop self-adaptive governance mechanisms for elastic privacy. Originally formulated for continuous IoT and personal sensor telemetry [28], the elastic privacy paradigm dynamically modulates the trade-off between privacy protection and sensing utility according to operational context. In vehicular opportunistic ISAC, this framework can be adapted to throttle sensing resolution based on vehicle motion state: maintaining fine-grained temporal resolution only at doorway thresholds during dwell stops to resolve boarding and alighting transitions, while dynamically coarsening spatial precision (e.g., binary cabin presence rather than continuous tracking) and adaptively quantizing latent representations during inter-stop cruising. Such context-driven elastic governance would guarantee verifiable privacy thresholds against longitudinal machine-learning profiling without compromising the macro-level counting fidelity required for fleet optimization. Models and code are available at [29].

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